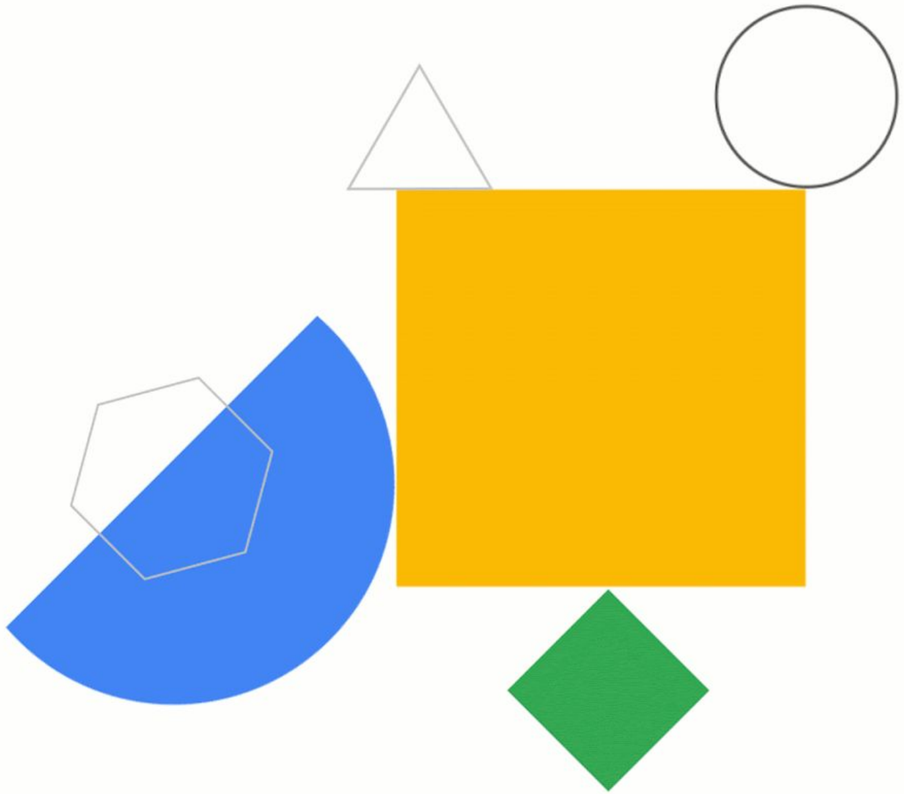
Google Cloud

Working with Cloud Audit Logs



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Objectives

01 Explain Cloud Audit Logs.

02 List and explain different audit logs.

03 Explain the features and functionalities of the different audit logs.

04 List the best practices to implement audit logs.



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In this section, you explore



- Cloud Audit Logs
- Data Access audit logs
- Audit logs entry format
- Best practices

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Cloud Audit Logs: “Who did what, where, and when?”

Admin Activity audit logs

Records modifications to configuration or metadata.

Helps answer questions such as “Who added that VM?”

System Event audit logs

Records Google Cloud non-human admin actions that modify configurations.

Helps answer questions such as “Did a live-migration event occur?”

Data Access audit logs

Records calls that read metadata, configurations, or that create, modify, or read user-provided data.

Helps answer questions such as “Who modified that Cloud Storage file?”

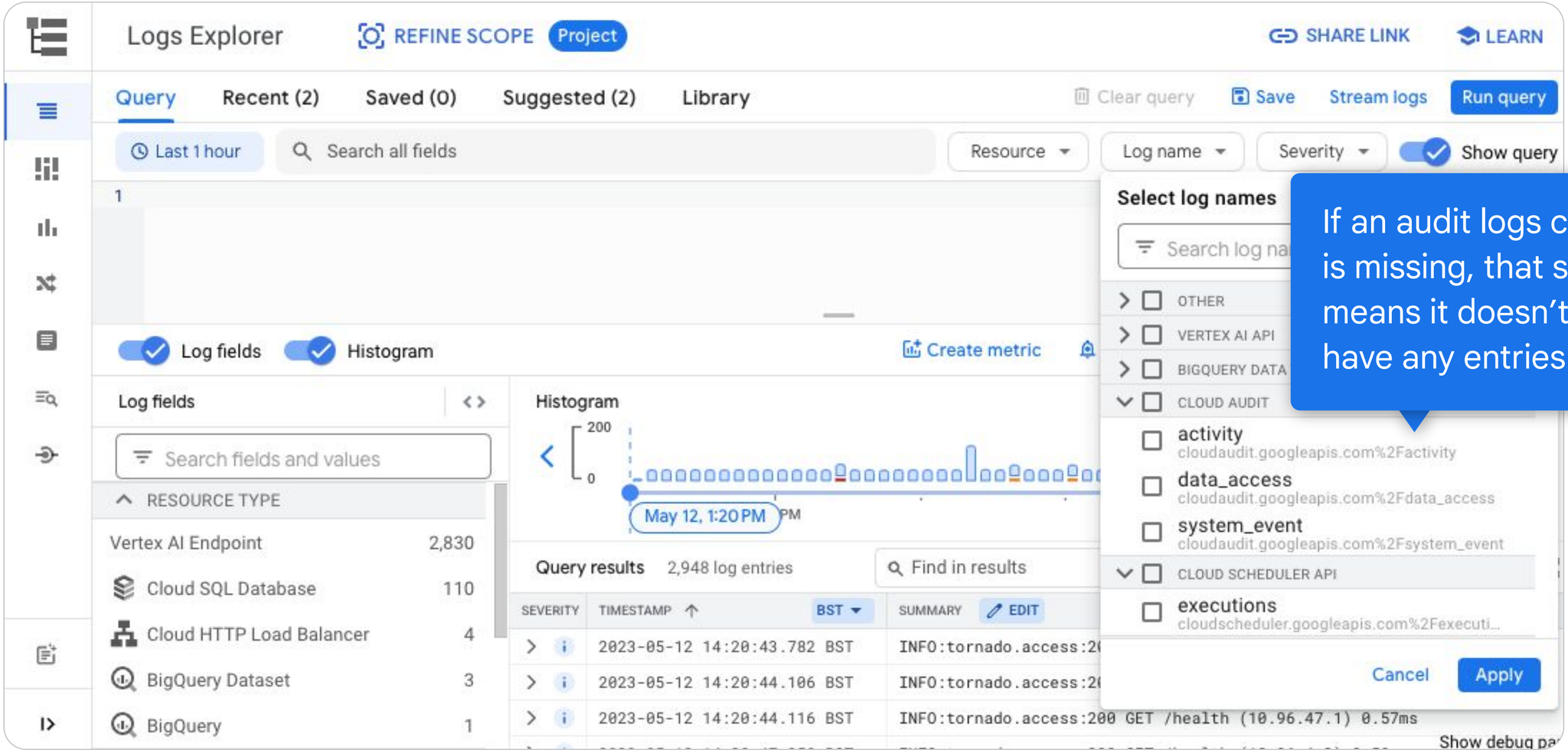
Policy Denied audit logs

Records a security policy violation.

Helps answer question such as “Who tried to breach a security policy?”

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Filtering audit logs, log names



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Access Transparency logs



Show **how** and **why** customer data is accessed once it has been stored in Google Cloud.

	Logs actions of accesses
	Tracks actions by Google personnel
	Supports approval and surfaced through App APIs and UIs, Security Command Center

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In this section, you explore

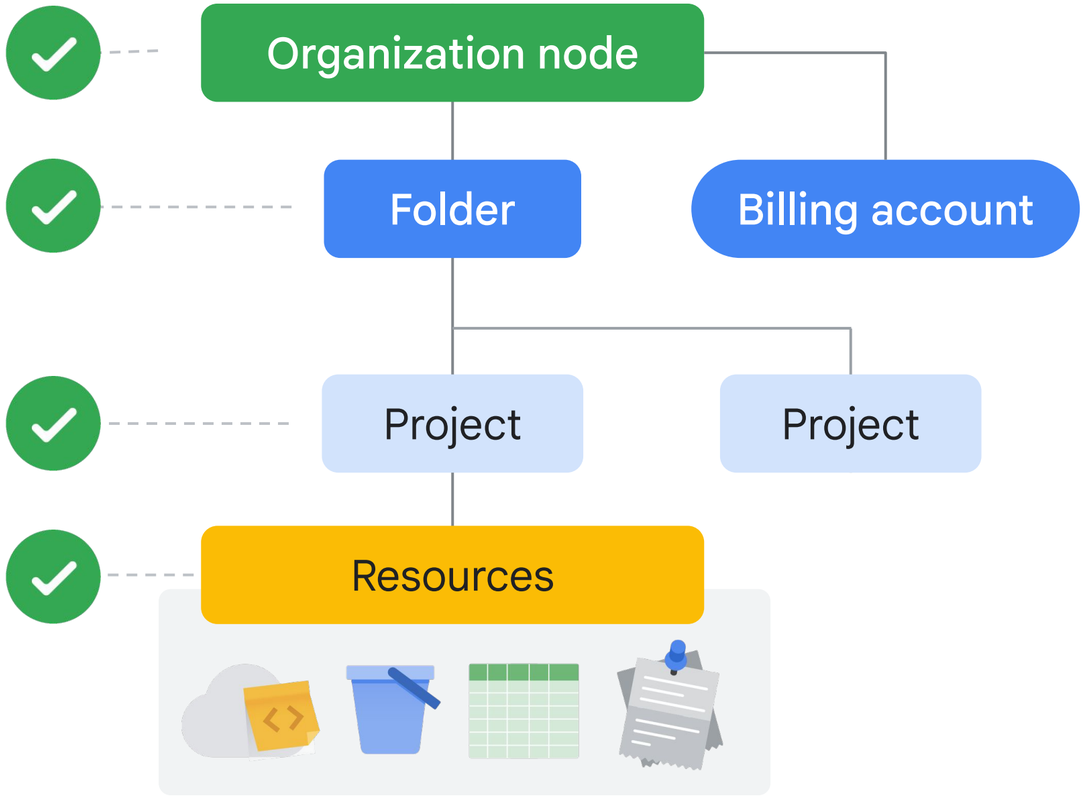


- Cloud Audit Logs
- **Data Access audit logs**
- Audit logs entry format
- Best practices

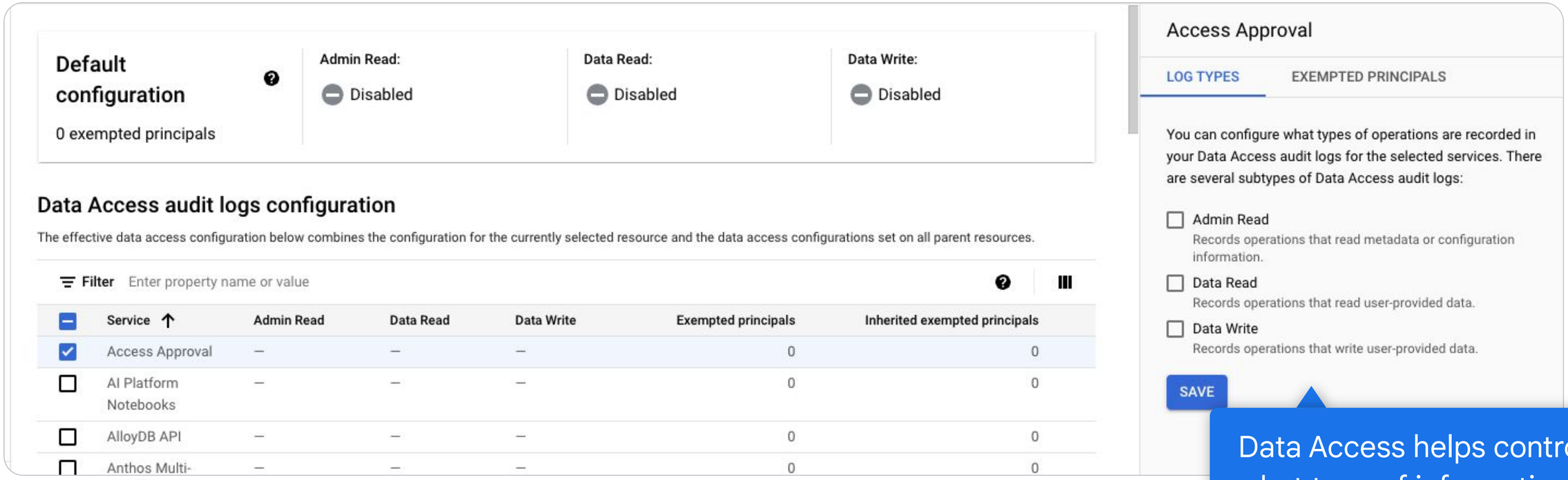
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Enabling Data Access audit logs in the organization

- Data Access audit logs can be enabled at:
  - Organization
  - Folder
  - Project
  - Resource
  - Billing accounts
- You can even exempt principals from recording data access logs
- Final scope is the union of the configurations



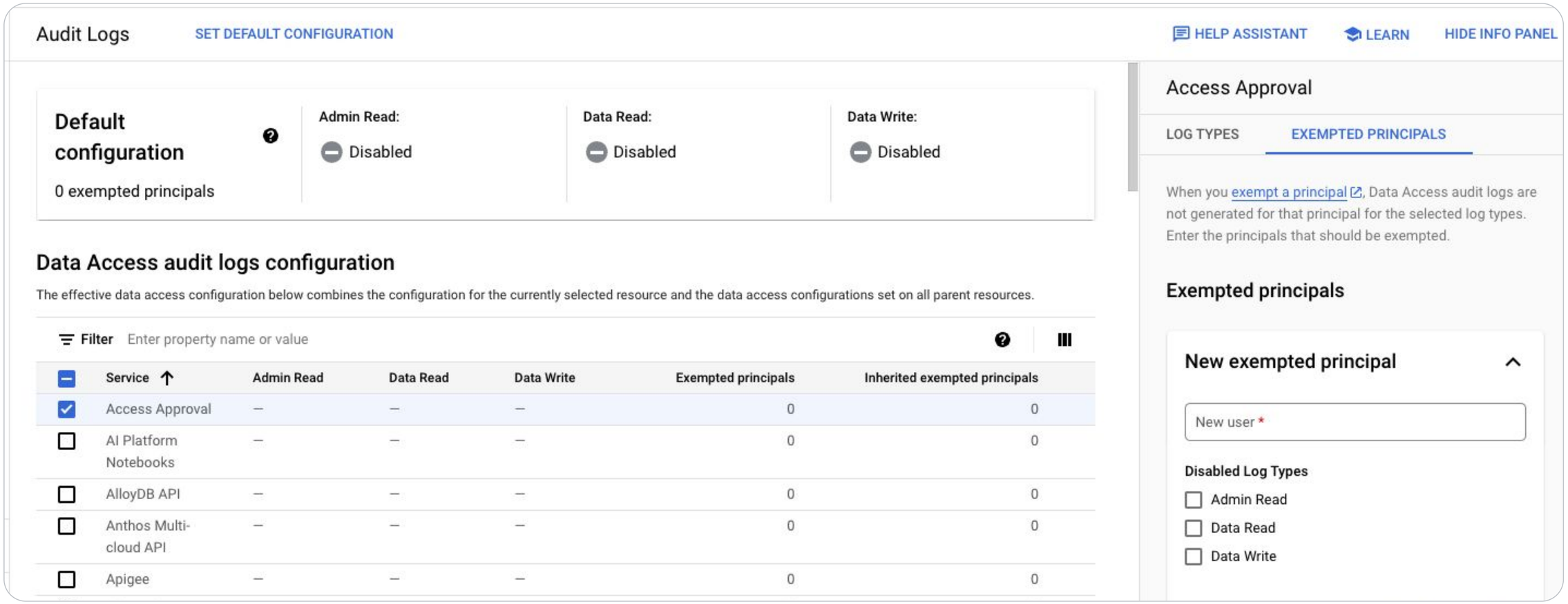
Enabling Data Access access logs per Google Cloud service



Data Access logs can be enabled and configured at the service level.

Data Access helps control what type of information is **recorded** in the Data Access audit logs.

Exempt specific users or groups



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Programmatically enabling Data Access audit logs

Step 1

```
gcloud projects get-iam-policy [project-id]  > policy.yaml
```

Step 2

```
auditConfigs:
- auditLogConfigs:
- logType: ADMIN_READ
  - logType: DATA_READ
  - logType: DATA_WRITE
  service: run.googleapis.com #Could also be all Services
bindings:
- members:
  ...
```

Step 3

```
gcloud projects set-iam-policy [project-id] policy.yaml
```

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In this section, you explore



- Cloud Audit Logs
- Data Access audit logs
- **Audit logs entry format**
- Best practices

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Cloud Audit Log entries

```
{
 insertId: "-77e5fge38tyo"
 logName: "projects/<projectID>/logs/cloudaudit.googleapis.com%2Fdata_access"
 operation: {
   first: true
   id: "1581200795118:bquxjob_56996f5_17026e67aa2"
   producer: "bigquery.googleapis.com"
 }

 protoPayload: {
   @type: "type.googleapis.com/google.cloud.audit.AuditLog"
   authenticationInfo: {
     principalEmail: "cloudysanfrancisco@gmail.com"
   }
```

Identify an audit log type

Identify the resource generating log by looking at **producer**

Identify an audit log entry by looking at the **protoPayload**

Identify the principal generating log by looking at the **principalEmail**

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Cloud Audit Log entries

```
 authorizationInfo: [2]
 methodName: "jobservice.getqueryresults"
 requestMetadata: {…}
 resourceName: "projects/<project_ID>/queries/bquxjob_1eb1f384_17026e9d185"
 serviceData: {…}
 serviceName: "bigquery.googleapis.com"
 status: {…} }
 receiveTimestamp: "2020-02-08T22:27:06.410866127Z"
 resource: {
  labels: { project_id: "<project_ID>", op_unit: "USA"
  }
  type: "bigquery_resource"
 }
 severity: "INFO"
 timestamp: "2020-02-08T22:27:05.908Z"
}
```

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In this section, you explore

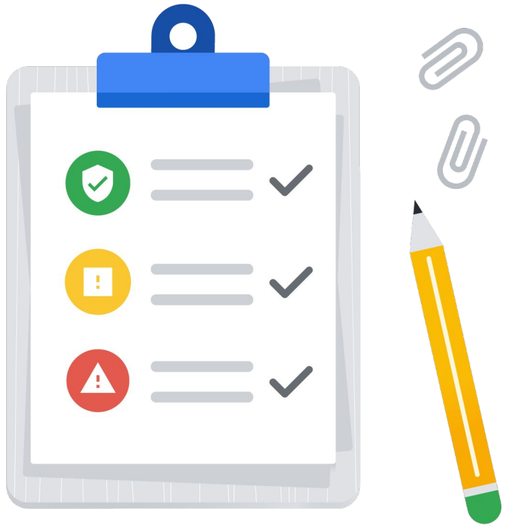


- Cloud Audit Logs
- Data Access audit logs
- Audit logs entry format
- **Best practices**

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Plan and create test project

- Create a plan for Data Access audit logs.
- Create a test project and test plan.
- Roll out the plan.



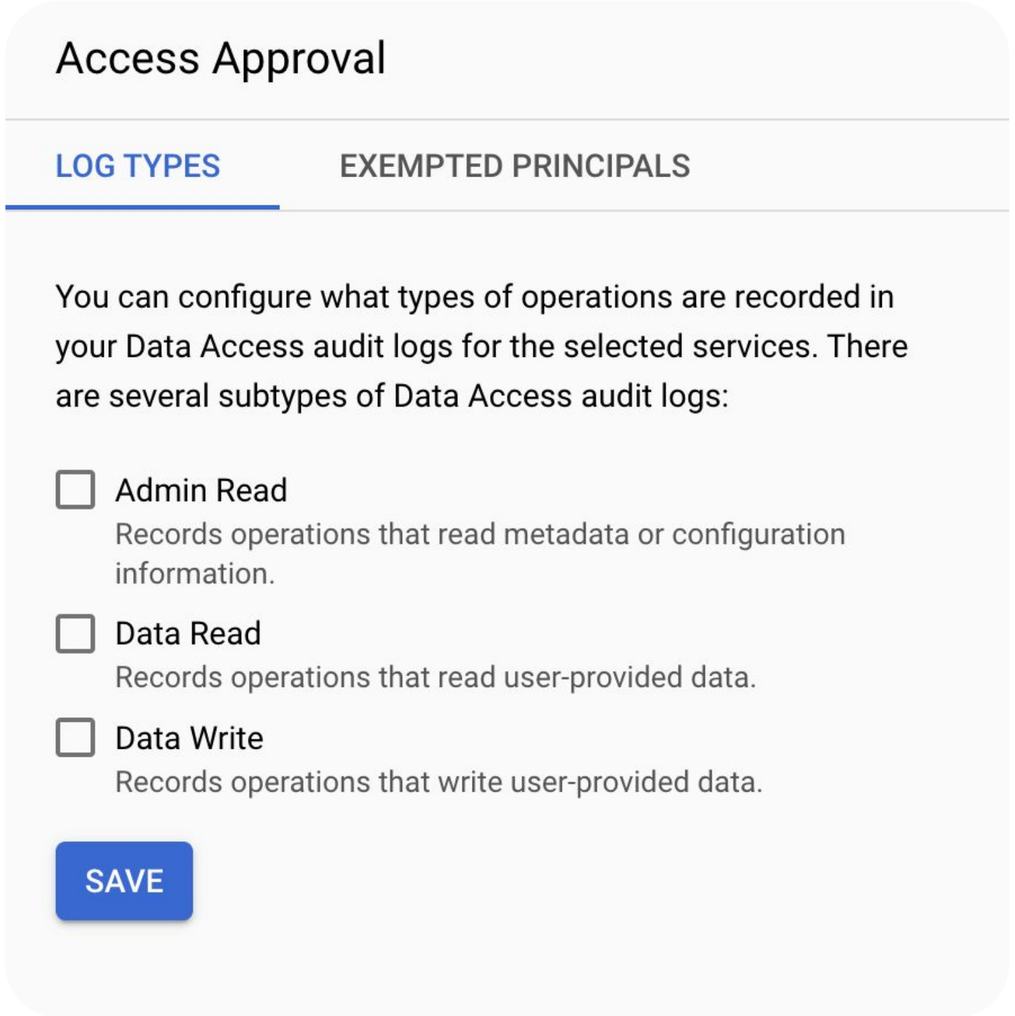
Decide and set org level data access

Advantages:

- Detailed information on who, accessed/edited/deleted what, and when
- Free tier
- Some logs are free

Disadvantage:

- Logs can be large and the Queries Per Second (QPS) can be high based on the number data access requests



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Infrastructure as Code (IaC)

Terraform: OSS or Enterprise

- Run OSS or pay for enterprise version.
- State can be stored locally or remote in Cloud Storage or Terraform Cloud.
- You can use Terraform to enable audit logs.
- Audit logs keep you informed of the resources provisioned using an IaC tool.

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Aggregate and store your organization's logs

- Centralize or subdivide log storage by creating user-defined buckets
- This helps meet latency, compliance and availability requirements
- Configure a default storage location at the organization level to automatically apply a region
- Protect your audit logs storage by configuring CMEK.

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Plan and configure exports




```
logName="projects/<project-id>/logs/cloudaudit.googleapis.com%2Fdata_access"
protoPayload.serviceName="bigquery.googleapis.com"
```

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Principle of least privilege

- Side-channel leakage of data through logs is a common issue.
- Plan the project to monitoring project relationships.
- Use appropriate IAM controls on both Google Cloud-based and exported logs.
- Data Access audit logs contain Personally Identifiable Information (PII).

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Configure log views

- Log views help control access to logs in a log bucket
- It helps control access specific to a project or a set of users
- It also help protect sensitive log data
- It ensures only authorized users have access to.

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Scenario: Operational monitoring

- CTO: **resourcemanager.organizationAdmin**
  - Assigns permissions to security team and service account.
- Security team: **logging.viewer**
  - Ability to view Admin Activity audit logs.
- Security team: **logging.privateLogViewer**
  - Ability to view Data Access audit logs.
- All permissions assigned at Org level.
- Control exported data access through Cloud Storage and BigQuery IAM roles.
- Explore using Sensitive Data Protection to **redact PII**.

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Scenario: Dev teams monitoring Audit Logs

- Security team, same:
  - **logging.viewer**, **logging.privateLogViewer**
- Dev team: **logging.viewer** at folder level
  - See Admin Activity audit logs by dev projects in folder.
- Dev team: l**ogging.privateLogViewer** at folder
  - See Data Access audit logs.
- Use Cloud Storage or BigQuery IAM to control access to exported logs
  - Providing a Dashboard might be helpful.

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Scenario: External auditors

- Provide Dashboards for auditor usage.
- **logging.viewer** at Org level
  - See Admin Activity audit logs by dev projects in folder.
- **bigquery.dataViewer** at exported dataset
  - Backend for Dashboards.
- For Cloud Storage, use IAM and/or, signed, temporary, URLs.

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Knowledge Check



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Quiz | Question 1

Question

Why are the Data Access audit logs off by default? Select THREE answers.

A. They can be large

B. They can be small

C. They may contain sensitive information

D. They can be expensive to store

E. They are formatted incorrectly

D. They cannot be filtered

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Quiz | Question 1

Answer

Why are the Data Access audit logs off by default? Select THREE answers.

A. They can be large 

B. They can be small

C. They may contain sensitive information 

D. They can be expensive to store 

E. They are formatted incorrectly

D. They cannot be filtered

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Quiz | Question 2

Question

If you want to provide an external auditor access to your logs, what IAM role would be best?

A. Project Viewer

B. Project Editor

C. Logging Viewer

D. Logging Admin

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Quiz | Question 2

Answer

If you want to provide an external auditor access to your logs, what IAM role would be best?

A. Project Viewer

B. Project Editor

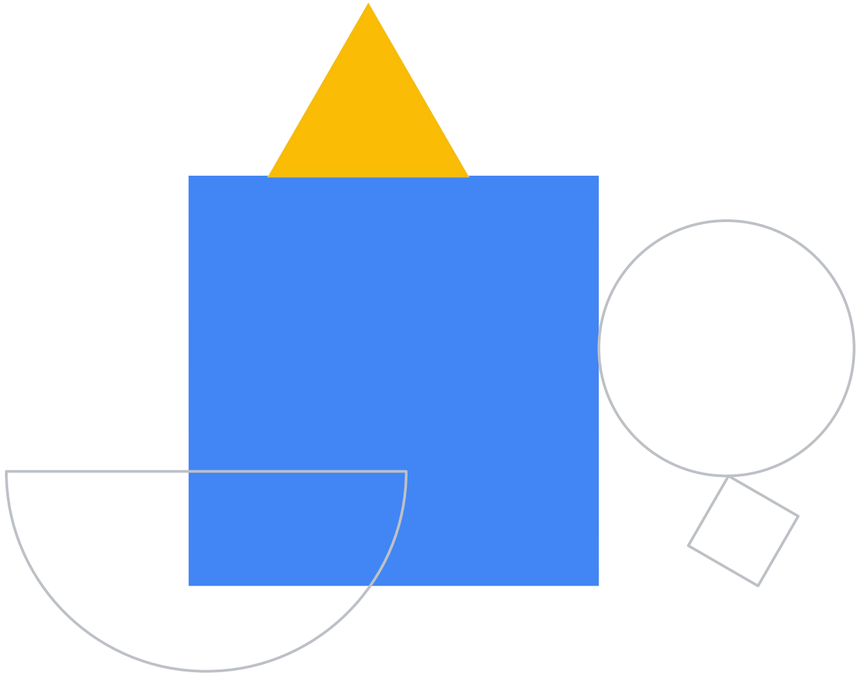
C. Logging Viewer 

D. Logging Admin

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Lab Intro

Cloud Audit Logs



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Recap

01 Explain Cloud Audit Logs.

02 List and explain different audit logs.

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